



MEDPAC PLATFORM OVERVIEW

MedPAC Software Description and Feature Scope

One governed imaging loop — DICOM as the source of truth, from the modality to the signed SR.

MedPAC is a complete medical imaging platform for modern hospitals, diagnostic centers, and multi-site healthcare networks. It combines a Vendor Neutral Archive (VNA), PACS workflow, universal medical image viewer, radiology reporting workspace, AI-assisted image analysis, cross-hospital DICOM exchange, compression and tiered storage, device communication gateways, and a central data/service broker into one connected system.

VNA / PACS

DICOMweb

IHE workflow

HL7 / FHIR

MedAI

Tiered storage

Device gateway

From edge to signed report, in one governed loop

For hospitals, IDNs, referral networks, radiology groups, and connected imaging operations.

1

Acquire from modality-adjacent PCs via PacMan, with durable store-and-forward.

2

Read, analyze, report, exchange, and audit in one provenance-preserving workspace.

3

Govern storage lifecycle, interoperability, security, and device policy centrally.

DATE

2026-06-12

PRODUCT

MedPAC

CATEGORY

Vendor Neutral Archive, PACS, radiology worktop, medical imaging AI, and hospital imaging workflow platform

Platform architecture, clinical workflow, the standards MedPAC is engineered around, and the full capability description — for technical, clinical, and procurement review.

[Platform overview](#)

Six properties that make it one platform, not a toolbox

The whole imaging lifecycle — acquisition, archive, reading, AI, reporting, exchange, and audit — behaves as a single governed system, with every artifact traceable to its source DICOM object.

01

One governed loop

Acquisition, archive, reading, AI, reporting, and exchange run on one UDB-projected schema, one identity model, and one audit trail — PostgreSQL and object storage behind every layer.

02

Provenance to the frame

Every measurement, DICOM SEG, GSPS, AI finding, and signed SR carries source Study / Series / SOP Instance / frame UUIDs plus model, user, timestamp, and tenant — re-exportable as DICOM and FHIR.

03

AI, radiologist in control

Image-type model gating, Grad-CAM grounding, and draft-only findings tagged with confidence, version, and provenance; deterministic Fleischner / Lung-RADS / RECIST engines — never free-text scoring.

04

Standards by design

DICOMweb (QIDO / WADO / STOW / UPS-RS), IHE workflow profiles, HL7 v2 plus FHIR R4 / R5, FHIRcast context sync, and ATNA audit — integration by named profile, not a bespoke bridge.

05

Lossless-first economics

Transfer-syntax-aware tiering across HTJ2K, JPEG 2000, and JPEG XL; hot-to-deep lifecycle, rules-based prior prefetch, and Object-Lock WORM with compression provenance on every derived object.

06

Durable at the edge

PacMan store-and-forward with bounded retry and DICOM Storage Commitment; a device gateway read-only by default with allowlists, dry-run, two-person approval, and tamper-evident audit.

The teams MedPAC is designed to serve

Hospitals & IDNs

A vendor-neutral archive and reading worktop unifying modalities, priors, AI, and reporting across departments and sites — one schema, one audit trail.

Imaging centers

Fast reads, structured SR output, and lossless-first storage economics without the weight or lock-in of a legacy PACS.

Referral & teleradiology

Governed second-opinion exchange with credential-aware assignment, SLA / TAT tracking, de-identified packaging, and full disclosure audit.

Research & AI programs

PS3.15 de-identification, exportable DICOM/FHIR evidence objects, and a model layer instrumented for versioning, monitoring, and governance.

Six cooperating planes, one operating layer

DICOM and DICOMweb remain the clinical source of truth; non-DICOM inputs are normalized; every plane reads and writes through the UDB-governed data layer.



One governed clinical imaging loop

Six planes share one proto-driven schema, one identity model, and one audit trail — so a study can be found, read, analyzed, reported, shared, and proven from a single source of truth.

01

PacMan Edge Node

Modality-adjacent acquisition



- Recursive watch folders, stable-file gating
- DICOM metadata extraction + dedup
- Durable store-and-forward queue, retry, heartbeat
- Local SQLite state + loopback control API

02

VNA / PACS Plane

Clinical source-of-truth archive



- Study / series / instance / frame identity
- DICOMweb QIDO · WADO · STOW · UPS-RS
- Object storage + Postgres metadata
- Tiered lifecycle, WORM, prior prefetch

03

MedView Worktop

Cornerstone3D radiology reading



- MPR / MIP / fusion, layouts up to 4×4
- Hanging-protocol rule engine, versioned
- Annotation, measurement, segmentation
- Structured reporting + sign-off lifecycle

04

MedAI Layer

Draft-only, human-in-control AI



- Vision: classify / detect / segment
- Cognition: DDx, VQA, draft narrative
- RADS / RECIST deterministic scoring
- Provider-agnostic serving contract

05

Exchange Fabric

Cross-enterprise second opinion



- Token-scoped DICOMweb to external orgs
- Referral packaging + de-identification
- Senior-reader queue, SLA, full audit
- FHIR ImagingStudy / DiagnosticReport projection

06

UDB + MedServe

Governed data + service plane



- Proto → schema → storage projection
- Double-entry migrations with checksums
- ABAC policy, CDC, saga coordination
- Generated Go / Python / C# SDKs

CLINICAL FLOW

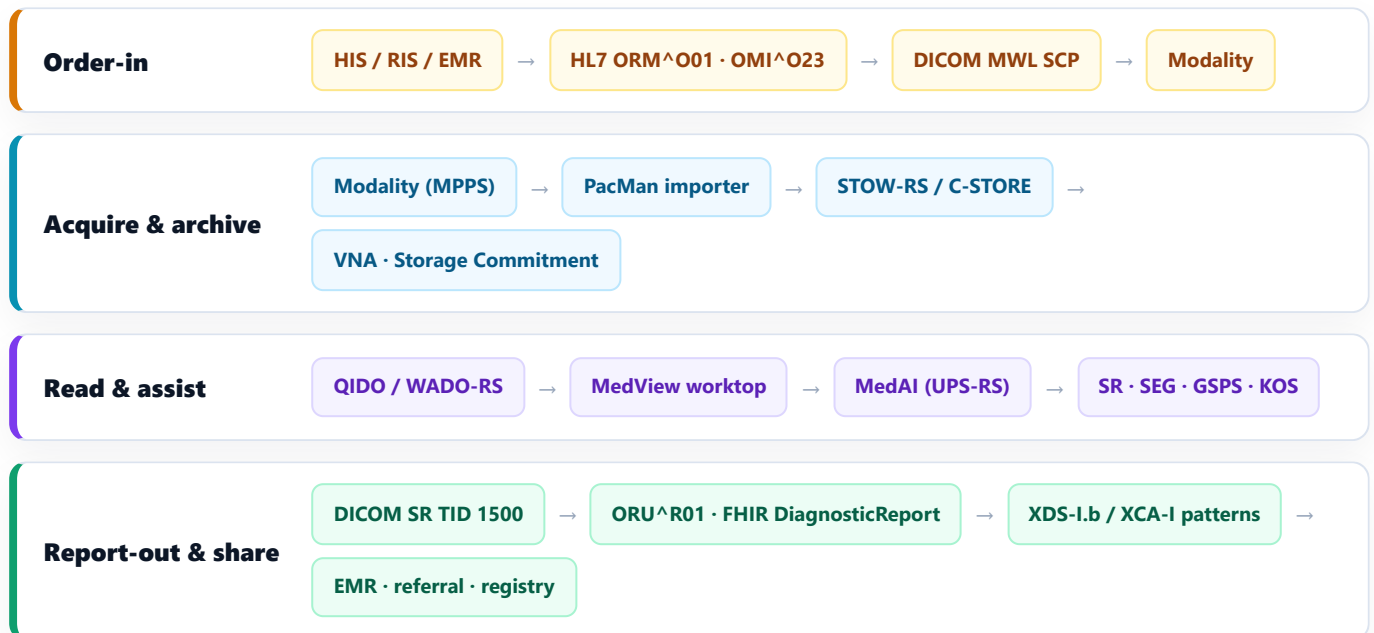
From machine room to signed report

A single, provenance-preserving path from acquisition to distribution — each step traceable to study, series, instance, model, user, and time.



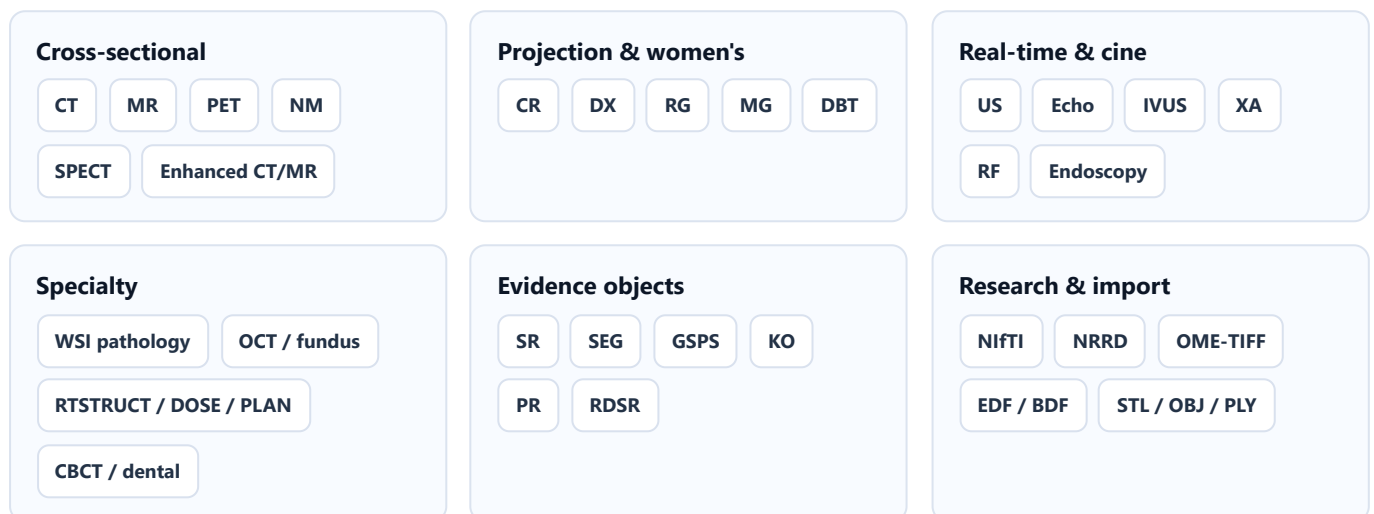
INTEROPERABILITY

Four data lanes, named transactions end to end



COVERAGE

What the universal viewer reads



Imaging & interoperability conformance

MedPAC is engineered around the imaging and interoperability standards enterprise healthcare procures by — declared with an honest posture, not a checkbox wall.

● Supported today
● Engineered around
○ Conformance-targeted

DICOM core & transfer syntax

- **DICOMweb services**
QIDO-RS · WADO-RS/URI · STOW-RS · UPS-RS (PS3.18)
- **DIMSE gateway**
C-STORE · C-FIND · C-MOVE · C-GET · MWL SCP · MPPS
- **Lossless syntaxes**
JPEG-LS · J2K · HTJ2K RPCL (...1.2.4.202) · JPEG XL (2024d) · RLE
- **Encapsulated video**
MPEG-4 AVC / HEVC (.102/.103/.107/.108) · Enhanced CT/MR
- **Conformance statement**
PS3.2, registry-generated AE / SOP / TS declarations
- **Display calibration**
PS3.14 GSDF QA · diagnostic-vs-review labeling

IHE Radiology profiles

- **Scheduled workflow**
SWF.b — Placer/Filler, MWL, MPPS, Storage Commitment
- **Presentation & key images**
CPI (GSPS) · KIN (KOS) round-trip
- **Change management**
IOCM — KOS rejection notes 113001/113037/113038/113039
- **Import reconciliation**
PIR · IRWF.b (PacMan as Importer, Original Attributes Seq.)
- **AI workflow & results**
AIW-I (UPS-RS workitems) · AIR (SR / SEG / parametric maps)
- **Web access & launch**
WIA · BIR · IID (RAD-106) · XDS-I.b / XCA-I

HL7 / FHIR interoperability

- **HL7 v2 messaging**
ADT · ORM^O01 · OMI^O23 · ORU^R01 · MLLP Enhanced ACK
- **FHIR resources**
ImagingStudy · DiagnosticReport · Observation · ServiceRequest
- **Context sync**
FHIRcast hub + subscribers (aligned to IHE IRA)
- **App launch**
SMART App Launch v2 + Backend Services (FHIR + DICOMweb scopes)
- **Patient identity**
PIXm (ITI-83/104) · PDQm (ITI-78/119) · Patient/\$match
- **Profiling & bulk**
US Core conformance · Bulk Data \$export

Structured reporting & terminology

- **SR content trees**
DICOM SR TID 1500 measurements · TID 2000/2001 narrative
- **Report templates**
IHE MRRT (RAD-103) · RSNA RadReport.org library
- **Coded terminology**
SNOMED CT · RadLex · LOINC/RPID · ICD-10-CM · CPT · UCUM
- **Structured assessments**
BI- / Lung- / LI- / PI- / TI- / O-RADS · RECIST 1.1 · Fleischner
- **Common data elements**
ACR / RSNA RadElement CDEs
- **Archival output**
PDF/A-3 + DICOM Encapsulated PDF · digital signature

Transaction-level conformance status is maintained in the MedPAC DICOM Conformance Statement (PS3.2) and IHE Integration Statement that accompany each deployment. Postures reflect the active development cycle; no FDA clearance or CE marking is implied.

Security, quality, and regulatory posture

The governance half of the conformance matrix — security and audit, plus the SaMD quality and AI-lifecycle frameworks the platform is being engineered against.

● Supported today ● Engineered around ● Conformance-targeted

Security, audit & governance

- **Audit of record**
DICOM PS3.15 + FHIR AuditEvent · TLS-syslog to ARR / SIEM
- **Cryptography**
FIPS 140-3 modules · HSM/KMS custody · AES-256 · TLS 1.3
- **Identity & access**
OIDC · SAML · MFA · IHE XUA (PurposeOfUse) · ABAC · break-glass
- **De-identification**
PS3.15 Annex E profiles + Clean Pixel Data (burned-in PHI)
- **Control mapping**
HIPAA Security Rule · NIST SP 800-66r2 / 800-53r5 / 800-207
- **Supply chain**
SBOM (SPDX / CycloneDX) · CVD policy · image signing

Quality, SaMD & AI lifecycle

- **Quality system**
ISO 13485 / FDA QMSR · 21 CFR 820.30 DHF + traceability
- **Risk & lifecycle**
ISO 14971 risk file · IEC 62304 safety class · IEC 62366-1
- **Regulatory pathway**
21 CFR 892.2050 · 510(k) / De Novo · Cures §3060 CDS criteria
- **AI change control**
FDA PCCP (Dec 2024) · GMLP · AI transparency labeling
- **AI surveillance**
ACR Assess-AI concordance · SPC drift (CUSUM) · model cards
- **European route**
EU MDR Rule 11 + Art. 61 CER/PMCF · EU AI Act Art. 17/19

Six security pillars in operation

Identity & trust

OIDC, SAML, MFA, service accounts, certificate-based node auth, and IHE XUA SAML assertions on cross-enterprise transactions.

Audit chain

Server-side append-only audit emitting DICOM PS3.15 and FHIR AuditEvent over TLS-syslog to a central repository and the hospital SIEM.

Crypto custody

AES-256 at rest, TLS 1.3 in transit, mTLS between services, envelope encryption with HSM/KMS key custody and audited rotation.

PHI minimization

PS3.15 Annex E de-identification with Clean Pixel Data, limited-data-set export, and a consent + data-use registry per referral route.

Device safety

Read-only telemetry by default; bounded, allowlisted, dry-run, two-person writes; emergency lockout; IEC 80001-1 + MDS2 posture.

Resilience

Replicated stateless services, synchronous database standby, quantified RPO / RTO targets, immutable backups, restore-validation drills.

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Final Product Description

Full capability description

The following pages describe each layer of the platform in detail, organized for clinical, technical, and procurement readers.



● Product Summary

MedPAC is a complete medical imaging platform for modern hospitals, diagnostic centers, and multi-site healthcare networks. It combines a Vendor Neutral Archive (VNA), PACS workflow, universal medical image viewer, radiology reporting workspace, AI-assisted image analysis, cross-hospital DICOM exchange, compression and tiered storage, device communication gateways, and a central data/service broker into one connected system.

The product starts at the hospital hardware and modality layer. Local imaging machines and workstations produce DICOM and other medical imaging files. A MedPAC site node, called **PacMan**, runs near those devices, watches configured folders or drives, extracts metadata, tags the image payload, deduplicates files, and sends studies securely to the central MedPAC storage and service plane. From there, MedPAC indexes, stores, displays, analyzes, reports, exports, and audits the full clinical imaging lifecycle.

The central user experience is **MedView**, a web-based PACS viewer and radiology worktop. MedView is designed so a radiologist can find a study, open it, review prior and current images, apply hanging protocols, measure and annotate, use segmentation tools, run AI-assisted analysis, generate a structured report, sign or amend the report, export clinical artifacts, and keep every action traceable to the source DICOM study, series, instance, and frame.

• Product Positioning

MedPAC is not only a viewer. It is an imaging operating layer for a hospital.

It is intended to provide:

1 A **VNA/PACS storage plane** for clinical image retention and retrieval.

2 A **site acquisition layer** that can run close to modalities and existing imaging PCs.

3 A **universal viewer** for DICOM, DICOMweb, NIFTI, browser images, research formats, pathology, waveforms, documents, video, meshes, and normalized proprietary inputs.

4 A **radiology reporting worktop** with templates, macros, dictation, measurement insertion, report lifecycle, exports, and audit history.

5 An **AI analysis engine** that routes images to suitable models, generates grounded findings, supports VQA, produces structured draft reports, verifies coding where possible, and keeps humans in control.

6 A **unified DICOM API** that can securely expose studies to approved external hospitals, referral centers, and senior radiologists for second opinion workflows.

7 A **compression and tiered storage engine** that manages DICOM transfer syntax conversion, cache policy, lifecycle movement, retention, archive cost, and retrieval speed.

8 A **machine communication gateway** that normalizes Modbus, CAN/CAN FD, serial, vendor-custom, and industrial protocol signals into guarded TCP/IP or ISO-on-TCP control channels for modality-adjacent communication and centrally governed device settings.

9 A **healthcare integration engine** for HL7 v2 ORM/OMI/ORU/ADT, DICOM Modality Worklist, XML, JSON, protobuf, FHIR, and site-specific message mappings.

10 A **service and data layer** based on MedServe microservices and UDB for schema-managed, tenant-aware, audited, high-performance access to PostgreSQL, object storage, model registry data, events, and cross-language SDKs.

● Platform Architecture

● 1. Facility Edge Layer: PacMan Sync Node

PacMan is the MedPAC Windows sync node and local hospital connector. It runs as a foreground process during development or as a Windows Service in production.

Current capability includes:

1 Recursive watch folders across one or more configured roots.

2 Stable-file detection before upload.

3 DICOM metadata extraction.

4 HTTP multipart upload to the MedPAC server.

5 Local SQLite upload state, retry, and deduplication.

6 SSE and polling job transport.

7 Heartbeats to the central server.

8 Loopback control API for health, status, queue inspection, service status, heartbeat, rescan, retry, ignore, and support-bundle actions.

9 Console and structured logging.

PacMan makes MedPAC deployable in real hospital environments where imaging devices may write to local folders, network shares, or modality-adjacent PCs before studies enter the central archive.

● 2. Central VNA and PACS Plane

The central MedPAC plane stores and indexes imaging data, metadata, reports, AI outputs, annotations, and workflow artifacts.

Core storage responsibilities include:

1 DICOM study, series, instance, and frame-level identity.

2 DICOMweb query, retrieve, and store flows using QIDO-RS, WADO-RS/WADO-URI, STOW-RS, and UPS-RS worklist/task support.

3 Object storage for image artifacts, AI masks, derived captures, exports, and model files.

4 PostgreSQL-backed metadata, lifecycle state, report status, audit records, and model registry data.

5 Sync jobs for PACS ingestion, AI inference, report distribution, and downstream workflow tasks.

MedPAC treats DICOM and DICOMweb as the clinical source of truth. Non-DICOM or proprietary formats are import sources and are normalized into standard clinical or research storage targets when needed.

● 3. MedView Diagnostic Worktop

MedView is the web-based imaging workspace. It is built around Cornerstone3D and the MedPAC viewer, reporting, AI, and enterprise workflow modules.

MedView is designed as one continuous reading workspace:

1 Worklist search and study selection.

2 PACS health visibility.

3 Multi-viewport diagnostic image review.

4 Hanging protocol selection and application.

5 Annotation, measurement, segmentation, and key image capture.

6 AI analysis review with heatmaps, labels, scoring, and draft findings.

7 Structured report authoring, dictation, signing, export, and amendment.

8 Audit trail for clinical and AI-assisted activity.

The product goal is that a study can be found, opened, read, annotated, measured, AI-assisted, reported, signed, exported, distributed, and audited inside MedView without losing provenance.

● 4. MedAI Analysis Layer

MedAI is the image analysis and AI orchestration layer. It connects deterministic vision models, hosted or local inference, medical VLM/LLM reasoning, structured scoring, coding verification, and MedView review surfaces.

The AI design uses a two-layer pattern:

1 A **vision layer** for classification, detection, segmentation, masks, heatmaps, measurements, and quantitative outputs.

2 A **cognition layer** for grounded findings, differential diagnosis, report draft text, VQA, comparison, patient/referrer summaries, and safety checks.

AI outputs remain draft clinical assistance. They are labeled, versioned, reviewable, editable, and auditable. The system does not auto-finalize a report.

● 5. MedServe and UDB Service Layer

MedServe provides the microservice surface for PACS, MedView, reporting, document generation, authentication, authorization, and enterprise integration.

UDB is the central data broker foundation. It is a Rust gRPC service and code generation layer that understands MedPAC proto definitions and projects them into storage, schema, SDK, and runtime behavior.

UDB capability includes:

1 Proto-based schema parsing and storage projection.

2 PostgreSQL table and catalog generation.

3 Foreign keys, indexes, vector-store projections, cache projections, object-store projections, model-registry projections, and multi-backend data targets.

4 Double-entry migration planning with proto and SQL ledger checksums.

5 Runtime system catalog for CDC, offsets, locks, ABAC policy, and saga coordination.

6 Read-replica routing, strong consistency routing, operation channels, backpressure, timeouts, metrics, and catalog hot reload.

7 Generated SDKs for Go, Python, and C# services.

The purpose of UDB is to prevent every microservice from inventing its own database access rules. It becomes the shared, tenant-aware, audited data access and schema-governance layer for the platform.

● 6. Unified DICOM API and Second Opinion Network

MedPAC provides a unified DICOM and DICOMweb API layer for hospitals that need to share imaging studies outside their own site. The same archive can support internal reading, external referral, remote senior-radiologist review, and second opinion workflows without copying data through unsafe ad hoc channels.

The second opinion surface includes:

1 Token-scoped DICOMweb access for approved external organizations.

2 QIDO-RS search, WADO-RS/WADO-URI retrieve, STOW-RS ingest, and metadata-only study access.

3 Study packaging for external review, including images, priors, reports, key images, measurements, AI outputs, and relevant clinical context.

4 De-identification or limited-data-set export policies per referral agreement.

5 Senior radiologist assignment, case queue, turnaround targets, acceptance, rejection, report/addendum return, and full audit trail.

6 FHIR ImagingStudy and DiagnosticReport projection so an outside EMR or referral portal can discover studies and retrieve reports through governed endpoints.

7 Cross-enterprise sharing alignment with IHE-style document and imaging exchange patterns such as XDS-I/XCA-I, ATNA audit, XUA authorization, and patient identity resolution through PIX/PDQ-style matching where needed.

This makes MedPAC usable as a regional imaging exchange layer, not just a single-hospital PACS.

● 7. Compression and Tiered Storage Engine

MedPAC includes a compression and storage lifecycle layer for controlling cost, retrieval performance, and standards-compliant image delivery.

The compression engine is responsible for:

- 1 Preserving the original received DICOM object as the legal/source artifact when required.
- 2 Producing derived transfer syntaxes for viewing, exchange, cold archive, and low-bandwidth remote review.
- 3 Supporting lossless-first clinical policy using DICOM transfer syntaxes such as Explicit/Implicit VR Little Endian, RLE, JPEG-LS lossless, JPEG 2000 lossless, HTJ2K lossless/RPCL, JPEG XL lossless, and Deflate where appropriate.
- 4 Supporting controlled lossy compression only under documented clinical, regulatory, modality, and body-part policies.
- 5 Tracking compression ratio, transfer syntax UID, codec version, source SOP Instance UID, derivation metadata, visual QA state, and who/what policy approved the conversion.
- 6 Serving rendered thumbnails, previews, lossy review copies, or full-fidelity diagnostic objects depending on user role, device, network, and clinical purpose.

The tiered storage service manages:

- 1 Hot cache for current worklists and active reads.
- 2 Warm storage for recent studies and frequently accessed priors.
- 3 Cold object storage for long-term archive.
- 4 Deep archive or offline tier for retention-heavy deployments.
- 5 Prefetch of relevant priors before reading sessions.
- 6 Object lock/WORM mode where policy requires immutable retention.
- 7 Lifecycle movement by modality, age, legal hold, report status, tenant, facility, payer, and clinical value.
- 8 Cross-region replication, checksum validation, repair jobs, and restore-status visibility in the viewer and API.

● 8. Machine Communication and Central Control Gateway

MedPAC can include a controlled machine communication gateway for modality-adjacent devices, acquisition PCs, sensors, and hospital equipment that expose industrial or vendor-specific interfaces.

The gateway normalizes:

1 Modbus RTU and Modbus TCP.

2 CAN and CAN FD device traffic where hardware adapters are available.

3 Serial RS-232/RS-485 device protocols.

4 Vendor-custom binary or text protocols.

5 OPC UA-compatible equipment models where the device or gateway supports it.

6 ISO-on-TCP / TCP/IP transport for centrally routed machine communication.

The purpose is to convert machine-level signals into a governed MedPAC device model for status, telemetry, configuration, and safe central control. It can read device state, expose acquisition room telemetry, manage allowed settings profiles, and coordinate with orders and modality worklists.

Because machine settings can affect patient safety and device operation, this layer must be policy-driven:

1 Read-only discovery and telemetry by default.

2 Explicit allowlists for writable settings.

3 Per-device command schema with units, bounds, prerequisites, and rollback behavior.

4 Two-person approval or local operator confirmation for high-risk commands.

5 Dry-run validation before writing settings.

6 Tamper-evident audit of every read, write, profile change, and central-control action.

7 Site-level emergency lockdown so local biomedical engineering can disable remote setting changes immediately.

● 9. HL7, XML, JSON, Protobuf, and DICOM MWL Integration Engine

MedPAC includes an integration engine for orders, demographics, scheduling, reports, billing hooks, and machine-readable service messages.

Supported integration forms include:

1 HL7 v2 ADT for patient and visit events.

2 HL7 v2 ORM^O01 and newer imaging order flows such as OMI^O23 for imaging service requests.

3 HL7 v2 ORU^R01 for report/result distribution.

4 DICOM Modality Worklist so acquisition devices receive patient, accession, requested procedure, scheduled procedure step, modality, protocol, station, and AE Title context before scanning.

5 DICOM Modality Performed Procedure Step and Instance Availability direction for acquisition completion and study availability workflows.

6 XML, JSON, and protobuf contracts for modern hospital systems, local adapters, and MedPAC microservices.

7 FHIR ServiceRequest, ImagingStudy, DiagnosticReport, Observation, Procedure, Endpoint, Task, and Communication mappings.

8 Site-specific mapping profiles for accession number, MRN, facility, department, provider, procedure code, body part, laterality, priority, transport mode, and billing identifiers.

This integration engine is what turns MedPAC from a viewer/archive into a hospital workflow system that can sit between HIS, EMR, RIS, modalities, PACS, AI services, reporting, and external referral networks.

● Core Feature Set

● Acquisition and Site Connectivity

1 Windows service sync node for hospital PCs and modality-adjacent workstations.

2 Configurable watch folders and multi-root folder monitoring.

3 DICOM metadata extraction and upload.

4 Local queue, deduplication, retry, and heartbeat.

5 Remote job handling through SSE or polling.

6 Loopback control API for local GUI and support tools.

7 Packaging surface for desktop dashboard, settings, logs, watch locations, installer, and support bundle export.

● VNA and PACS Workflow

- 1 DICOM study, series, and instance indexing.
- 2 DICOMweb QIDO, WADO, STOW, and UPS client support.
- 3 Unified DICOM API for internal clients, external hospitals, referral portals, and approved second opinion readers.
- 4 Local DICOM Part 10 loading.
- 5 Metadata-first retrieval and pixel streaming.
- 6 Imported object upload for SEG, SR, PR, screenshots, and AI outputs.
- 7 PACS health monitoring and worklist integration.
- 8 Report status visibility from the worklist.
- 9 Sync jobs for AI inference and report distribution.

● Second Opinion and Cross-Hospital Exchange

- 1 Secure external organization onboarding with tenant, facility, department, and reader-level access boundaries.
- 2 Case referral package containing current study, priors, reports, clinical indication, key images, measurements, AI summaries, and requested question.
- 3 DICOMweb and FHIR endpoints for outside hospitals that want API-based integration instead of portal-only access.
- 4 Senior radiologist queue with assignment, SLA, priority, acceptance, hold, completed, amended, and returned states.
- 5 De-identification, limited-data-set, consent, and data-use policy per referral route.
- 6 Watermarking and viewer restrictions for external review sessions where required.
- 7 Report return as structured MedPAC report, DICOM SR, PDF, HL7 ORU, FHIR DiagnosticReport, JSON, or protobuf event.
- 8 Complete audit trail covering who shared the study, who opened it, which images were viewed, which report was returned, and whether the original institution accepted it.

● Compression and Tiered Storage

- 1 DICOM transfer syntax inspection, negotiation, and conversion.
- 2 Lossless-first compression policy for diagnostic archive and exchange.
- 3 Controlled lossy compression for non-diagnostic previews, web thumbnails, low-bandwidth review, and clinically approved use cases.
- 4 Codec support policy for JPEG, JPEG Lossless, JPEG-LS, JPEG 2000, HTJ2K, JPEG XL, RLE, Deflate, MPEG2, H.264, H.265, and encapsulated uncompressed DICOM paths.
- 5 Compression provenance with source SOP UID, derived SOP UID, transfer syntax UID, codec, ratio, quality policy, and validation status.
- 6 Hot, warm, cold, and deep-archive object tiers.
- 7 Lifecycle movement by study age, modality, report status, facility, tenant, legal hold, access frequency, and retention policy.
- 8 Prior-study prefetch, thumbnail generation, edge cache, restore queue, checksum validation, and archive repair workflow.

● Machine-Level Communication and Central Control

- 1 Common device gateway for Modbus RTU/TCP, CAN/CAN FD, serial, OPC UA, vendor-custom protocols, and ISO-on-TCP/TCP conversion.
- 2 Normalized equipment registry for modality-adjacent devices, acquisition PCs, sensors, and room-level equipment.
- 3 Telemetry collection for status, uptime, error codes, environmental readings, queue state, and acquisition readiness.
- 4 Centrally governed settings profiles for devices that support safe remote configuration.
- 5 Command allowlists, bounds checking, units validation, dry-run mode, local operator confirmation, approval workflow, and emergency lockout.
- 6 Audit trail for every device read, write, profile push, failed command, rollback, and manual override.
- 7 Integration with DICOM MWL and order context so modality settings can be checked against scheduled procedure, protocol, body part, station, and patient safety rules.

● Enterprise Integration Engine

- 1 HL7 v2 ADT, ORM^O01, OMI^O23, ORU^R01, ACK/NACK, order status, cancellation, replacement, and report-result flows.
- 2 DICOM Modality Worklist generation from orders and scheduling data.
- 3 Procedure and protocol mapping from HL7 orders to DICOM MWL fields including accession, requested procedure, scheduled procedure step, modality, protocol code, station name, location, and AE Title.
- 4 XML, JSON, and protobuf message transformation for modern hospital APIs and MedPAC service-to-service contracts.
- 5 Message validation, replay, dead-letter queue, retry, idempotency, correlation ID, and interface-level audit logs.
- 6 Site-specific mapping profiles with test harnesses and sample-message fixtures for each hospital integration.

● Universal Medical Image Viewer

MedView recognizes and routes a broad range of clinical, research, proprietary, and derived image formats.

Supported or covered areas include:

- 1 CT, MR, PET, NM, SPECT, CR, DX, RG, mammography, DBT, ultrasound, echo, IVUS, angiography, fluoroscopy, radiation oncology, ophthalmology, pathology, endoscopy, visible-light imaging, dental/CBCT, waveforms, documents, SR, KO, PR, SEG, and RT objects.
- 2 DICOM, DICOMweb, browser raster images, NIfTI, NRRD, TIFF/BigTIFF/OME-TIFF paths, plain video, PDF, EDF/BDF waveform, DICOM waveform preview, STL/OBJ/PLY mesh preview, and normalized import routes for formats that need server-side conversion.
- 3 Capability registry that records runtime support, server requirements, license requirements, and normalization targets.
- 4 Provenance records for loaded source images and derived objects.
- 5 Safe routing for unsupported, vendor-licensed, and server-import-only inputs.

● Diagnostic Viewing

- 1 Cornerstone3D stack viewport.
- 2 Window/level, pan, zoom, rotate, flip, invert, reset.
- 3 Cine-style stack review hooks.
- 4 Multi-viewport layouts up to 4x4.
- 5 MPR, oblique MPR, MIP, MinIP, AIP, and derived volume display modes.
- 6 PET/CT-style fusion stack generation.
- 7 Linked scrolling, camera, VOI, zoom/pan sync groups.
- 8 Reference lines, localizers, orientation markers, overlays, rulers, scale readouts, HU/SUV/intensity probe, and modality VOI presets.
- 9 Viewport snapshots with provenance.
- 10 Presentation state and key-object export packages.
- 11 Multi-monitor/layout preferences and touch/tablet support.
- 12 Keyboard workflow support for reading sessions.

● Hanging Protocols

- 1 Local and enterprise hanging protocol rule engine.
- 2 Built-in protocols for common modalities and body parts.
- 3 Custom protocol editor.
- 4 User, group, site, and system-scoped protocols.
- 5 Versioning, approval workflow, clone history, rollback, import/export, and simulation.
- 6 Required and optional matching rules with scoring.
- 7 Prior-study matching, longitudinal layouts, multi-stage protocols, viewport type rules, sync groups, and initial viewport rules.
- 8 Audit events when protocols are applied manually, automatically, by rollback, or by stage change.

● Annotation and Measurement

- 1 Cornerstone annotation summaries.
- 2 Server-adapter-backed persistence with local fallback.
- 3 JSON import/export and Cornerstone rehydration.
- 4 Patient-space and frame-of-reference-aware annotation storage.
- 5 Length, bidirectional, angle, Cobb, probe, arrow, text, rectangle ROI, ellipse ROI, freehand ROI, spline ROI, and livewire contour tools.
- 6 Manual calibration for non-calibrated X-ray/photo images.
- 7 Measurement statistics including length, perimeter, area, min, max, mean, standard deviation, volume, HU, SUV, and related quantitative values where available.
- 8 Lesion IDs, target/non-target grouping, baseline/current/prior comparison, and RECIST-style workflows.
- 9 Lock, visibility, label, color, grouping, author, timestamp, approval state, filters, undo/redo, QA validation, and audit events.
- 10 Export contracts for DICOM SR, GSPS, and KO.

● Segmentation and Advanced Visualization

- 1 Local mask layers and Cornerstone segmentation integration.
- 2 Labelmap, contour, and surface segmentation representations.
- 3 Conversion between labelmap, contour, and surface objects.
- 4 Stack and volume labelmaps.
- 5 Multi-segment labelmaps with colors, labels, opacity, visibility, lock state, active segment, and statistics.
- 6 Brush, erase, scissors, threshold, island removal, smoothing, interpolation, fill holes, logical union/intersection/subtract, livewire, spline, and freehand contour workflows.
- 7 2D and 3D segmentation visualization.
- 8 DICOM SEG import/export contract, RTSTRUCT import/export contract, NIfTI labelmap import/export, and STL/OBJ/PLY/VTP surface export contract.
- 9 AI-assisted segmentation gateway with promptable segmentation, draft masks, confidence, model name/version, timestamp, and review/accept/reject flow.
- 10 Active learning package metadata with consent and provenance controls.

● Specialty Imaging Workflows

- 1 Digital pathology WSI workflow surfaces, tile viewport direction, minimap, magnification, scalebar, stain/channel controls, annotations, and heatmap-ready structures.
- 2 Ophthalmology OCT/fundus workflow support with progression review concepts.
- 3 Mammography CC/MLO pairing, laterality-aware hanging, linked zoom/pan, prior comparison, and CAD/SR mark handling.
- 4 Radiation oncology CT, RTSTRUCT, RTDOSE, RTPLAN workflow coverage, dose colorwash direction, and DVH-ready export data.
- 5 Cardiology echo/angiography cine workflow, frame timing, stenosis tools, and ECG synchronization direction.
- 6 Ultrasound cine, Doppler metadata, OB and echo measurement support.
- 7 Dental/CBCT, endoscopy, dermoscopy, wound, visible-light, mesh, waveform, and document review surfaces.

● AI Image Analysis

The AI layer is built to provide clinically useful assistance while keeping the radiologist in control.

Current and release-scope capability includes:

- 1 Health and versioned model status for the hosted inference backend.
- 2 Image type detection and model gating so the CXR model is not run blindly on non-CXR studies.
- 3 Chest X-ray multi-label classification using TorchXRayVision DenseNet-121 via the MedPAC CNN service.
- 4 Reliable-label filtering to reduce false alarms on normal studies.
- 5 Pneumonia and abnormal CXR detection grounded into the LLM report.
- 6 Cardiothoracic ratio measurement for frontal CXR.
- 7 Grad-CAM heatmap overlay for reliable positive CXR findings.
- 8 Routed classifier support for bone fracture, skin lesion, and diabetic retinopathy tasks.
- 9 Interactive VQA over uploaded images.
- 10 Differential diagnosis, structured scoring, suggested ICD-10-CM codes, and ICD-10 verification.
- 11 Longitudinal prior-vs-current image comparison.
- 12 Burned-in PHI detection and PHI risk banner.
- 13 Structured per-finding objects with label, laterality, location, severity, confidence, and provenance.
- 14 Deterministic scoring calculators for Fleischner 2017, Lung-RADS v2022, RECIST 1.1, with TI-RADS and BI-RADS as additional clinical scoring targets.
- 15 Prediction-stability indicators for CXR model outputs.
- 16 Patient and referrer summaries.
- 17 AI triage scoring, STAT/criticality badges, critical findings acknowledgement, closed-loop audit, and worklist priority sorting.
- 18 MedSAM promptable segmentation from bounding box prompt to draft mask.
- 19 FHIR R4 export of analysis results as DiagnosticReport, Observation, and ImagingStudy-linked bundles.

The AI serving layer is intentionally provider-agnostic. The same grounding contract can come from a HuggingFace Docker Space, a dedicated endpoint, or local ONNX/gRPC inference, so MedView can use hosted or on-premises inference without changing the review workflow.

● Reporting and Report Distribution

MedView includes a client-first reporting workspace and report persistence model.

Current capability includes:

- 1 Structured **RadiologyReport**, report sections, findings, templates, status, version snapshots, audit events, and pluggable server adapter.
- 2 Report lifecycle: draft, preliminary, final, addendum/amended.
- 3 Report creation, template application, transition guard, persistence, load latest, and report listing.
- 4 Local fallback storage and REST adapter path for server integration.
- 5 Template library for generic, CT chest, CT abdomen/pelvis, chest X-ray, MR brain, and ultrasound abdomen reports.
- 6 Section editor for Technique, Comparison, Findings, Impression, and Recommendations.
- 7 Measurement insertion from viewer annotations.
- 8 Key image insertion from the active viewport.
- 9 Findings list with remove and insert-into-report actions.
- 10 Save draft, mark preliminary, finalize with required-field gating, and amend/addendum flow.
- 11 Export to printable HTML/PDF, text, JSON, and DICOM SR contract payload.
- 12 Version history and audit events for save, finalize, amend, and export.
- 13 Web Speech API dictation behind a pluggable transcription adapter, with Groq Whisper support through a server-side proxy path.
- 14 Macros, normal boilerplate insertion, smart impression generation, and comparison text from longitudinal measurement tracking.
- 15 Worklist report-status visibility and direct report entry point.

The reporting product boundary is clear: operational business reports remain separate from radiology narrative reports. Radiology reports belong to the MedView/reporting workflow and must preserve source image and measurement provenance.

● Interoperability and Standards

MedPAC is designed around clinical interoperability.

Included standards and integration surfaces:

1 DICOM Part 10.

2 DICOM DIMSE gateway capability for C-STORE, C-FIND, C-MOVE, C-GET, Modality Worklist SCP, and Modality Performed Procedure Step integration where hospital equipment does not yet support DICOMweb.

3 DICOMweb QIDO-RS, WADO-RS/WADO-URI, STOW-RS, UPS-RS.

4 DICOM Modality Worklist and modality workflow context.

5 DICOM transfer syntax policy for native, encapsulated, compressed, video, waveform, rendered, and metadata-only access.

6 DICOM SR, DICOM SEG, GSPS, KO, RTSTRUCT, RTDOSE, RTPLAN, Encapsulated PDF, and secondary capture export contracts.

7 FHIR R4 DiagnosticReport, Observation, and ImagingStudy bundles.

8 HL7 v2 ADT, ORM, OMI, ORU, ACK/NACK, order-control, and imaging-procedure-control flows.

9 HL7 CDA and FHIR-based report distribution.

10 XML, JSON, and protobuf message contracts.

11 XDS-I/XCA-I-style image sharing, enterprise worklist, export, quarantine, launch, search, and interop contract registries.

12 Cloud DICOM source configuration for AWS HealthImaging, Google Cloud Healthcare API, and Azure Health Data Services.

13 BUF-based proto definitions and generated clients for Go, Python, TypeScript, C#, and other service surfaces.

● Enterprise Standards, Exchange, and Governance Features

MedPAC presents enterprise standards as platform capabilities that hospitals can directly use across clinical workflow, infrastructure, security, and connected-device operations.

1 DICOMweb study search, metadata, retrieval, and storage for browser viewers, external integrations, second-opinion exchange, and downstream clinical systems.

2 FHIR ImagingStudy projection for study, series, instance, endpoint, and report linkage while DICOM/DICOMweb remains the image-object retrieval plane.

3 DICOM Modality Worklist generation from HL7 order data, including accession, requested procedure, Study Instance UID, scheduled procedure step, modality, protocol, station, location, and AE Title.

4 Structured report/result exchange through FHIR DiagnosticReport, Observation, presented PDF, DICOM SR, HL7 ORU, JSON, XML, and protobuf events.

5 Compression-aware retrieval for large CT, MR, DBT, pathology, and video-heavy studies using transfer syntax policy, targeted retrieval, cache, prefetch, and tiered storage.

6 Cross-enterprise image sharing using external organization boundaries, secure referral packaging, identity reconciliation, audit, consent, and second-opinion report return.

7 Enterprise imaging workflow coverage for scheduled workflow, patient reconciliation, consistent presentation, key images, reporting workflow, evidence documents, acquisition protocol management, remote reading, AI workflow, AI results, and operational event logs.

8 Secure node and secure application behavior with user authentication, machine authentication, access control, certificate-based trust, encrypted transport, audit repositories, audit forwarding, and immutable audit retention.

9 HIPAA-oriented ePHI safeguards covering confidentiality, integrity, availability, risk analysis, access control, audit controls, authentication, transmission security, contingency planning, documentation, and external-organization agreements.

10 Medical-device cybersecurity controls for the machine communication gateway, including bounded writes, explicit approval, local override, rollback, auditability, and resilient operation.

11 OPC UA-style device modeling for secure channels, address spaces, methods, subscriptions, alarms/events, XML/JSON/binary encodings, TCP/HTTPS/WebSocket transports, and publish/subscribe telemetry patterns where supported by devices or adapters.

● Security, Governance, and Compliance

MedPAC handles protected health information and clinical workflow artifacts, so security is part of the product surface.

Required and covered areas include:

1 Tenant-aware routing and role/scope-aware access control.

2 Authentication and authorization contracts, including OIDC, SAML, MFA, service accounts, machine identities, API keys, token scopes, and certificate-based trust.

3 PHI detection and de-identification controls.

4 Tamper-evident audit chain contracts.

5 ATNA-style audit events for login, logout, failed access, node authentication, query, retrieve, store, export, report transition, device-control action, AI run, critical finding acknowledgement, FHIR export, and external sharing.

6 Audit repository, audit forwarding, SIEM/syslog export, correlation IDs, time synchronization, and immutable log retention.

7 Encryption in transit and at rest, mTLS for service-to-service calls, certificate rotation, secrets management, key-management integration, and secure backup encryption.

8 HIPAA-oriented administrative, physical, and technical safeguard evidence: risk analysis, access review, audit review, incident response, workforce access control, contingency plan, backup/restore testing, and documentation retention.

9 Business-associate, data-processing, data-use, referral, and second-opinion agreement tracking for external organizations.

10 SBOM, vulnerability intake, dependency scanning, image signing, patch policy, threat model, penetration-test evidence, and secure software update workflow.

11 Break-glass access with reason capture, short-lived elevated session, notification, and after-action review.

12 Consent, minimum-necessary access, de-identification, limited-data-set export, research export, and external sharing policy registry.

13 Compliance evidence export and audit verification surfaces.

14 Human-in-the-loop AI framing and model provenance.

● Current Development-Cycle Capability Statement

This document describes MedPAC as the capability target for the current active development cycle. The implementation already contains the major product modules and the end-of-month cycle is intended to make those modules read as one connected platform:

1 PacMan connects hospital-side folders and modality PCs to the central MedPAC plane.

2 The PACS/VNA layer indexes studies, exposes DICOMweb flows, and supports sync jobs.

3 MedView provides the diagnostic reading workspace, worklist, viewer, hanging protocols, annotation, measurement, segmentation, AI review, and reporting UI.

4 MedAI provides image analysis, CNN/VLM/LLM grounding, heatmaps, scoring, coding support, comparison, PHI risk checks, triage, and FHIR export.

5 UDB and MedServe provide the backend service, schema, migration, data access, catalog, SDK, and runtime foundation.

6 The unified DICOM API, second-opinion exchange, DICOM MWL/order integration, HL7/XML/JSON/protobuf integration engine, compression/tiered storage service, and machine communication gateway are included in the enterprise capability scope for the same active cycle.

7 Interoperability and security contracts are present so clinical exports, audit, access control, and compliance evidence can be connected consistently.

The product should therefore be described as an integrated hospital imaging platform whose components are already being completed in the current engineering cycle.

● Final Product Description

MedPAC is a hospital-grade VNA, PACS, AI analysis, and reporting platform. It connects local imaging sources through PacMan, stores and indexes studies in a central DICOM-aware service plane, gives radiologists a full MedView diagnostic worktop, adds AI-assisted analysis and triage through MedAI, and uses MedServe plus UDB as the governed service and data foundation.

The platform is built so every clinical artifact remains traceable: an uploaded image, a measurement, a segmentation, an AI finding, a report section, a signed report, a FHIR export, and an audit event can all point back to the original study, series, instance, frame, model, user, timestamp, and tenant context. That traceability is the core value of MedPAC as a modern hospital imaging system.

With the unified DICOM API, second-opinion exchange, DICOM MWL and HL7 integration engine, compression and tiered storage service, and guarded machine communication gateway, MedPAC is positioned as an enterprise imaging operating layer for connected hospitals, referral networks, senior radiology review, clinical AI, storage governance, and safe modality-adjacent operations.